Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	27-10-2026
Team ID	_____
Project Name	Hotel Pro Booking and Management System
Maximum Marks	4 Marks

Technical Architecture:

The Hotel Pro Booking and Management System is designed with a simple 3-tier architecture. It includes the presentation layer (frontend), application layer (business logic), and data storage layer (database). The system helps guests search rooms, check availability, make bookings and payments, while hotel staff manage rooms, reservations and guest records.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web pages for guests and hotel staff	HTML, CSS, JavaScript
2	Application Logic	Room search, availability, booking and cancellation	Python Flask / Node.js
3	Booking & Payment	Reservation confirmation and payment status	REST API, payment gateway API
4	Admin Panel	Manage rooms, rates, bookings and guest details	HTML, CSS, JavaScript
5	Database	Stores users, rooms, bookings and payment records	MySQL

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Tools	Tools and frameworks used to develop the system	HTML, CSS, Flask, MySQL
2	Scalable Architecture	Separates interface, application logic and database	3-tier architecture
3	Security	Protects accounts and booking/payment data	Password hashing, HTTPS

References:

1. HTML and CSS Documentation
2. Flask Documentation
3. MySQL Documentation
4. REST API design references